
A

Summary Report On

Road Safety Analysis Dashboard

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PowerBi DashBoard

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1 Objective

To develop an interactive Power BI dashboard analyzing road accident data, with the goal of identifying trends, environmental impacts, and severity factors contributing to accidents. This submission showcases data preparation, exploratory analysis, and visual storytelling through Power BI.

2 Data Preparation

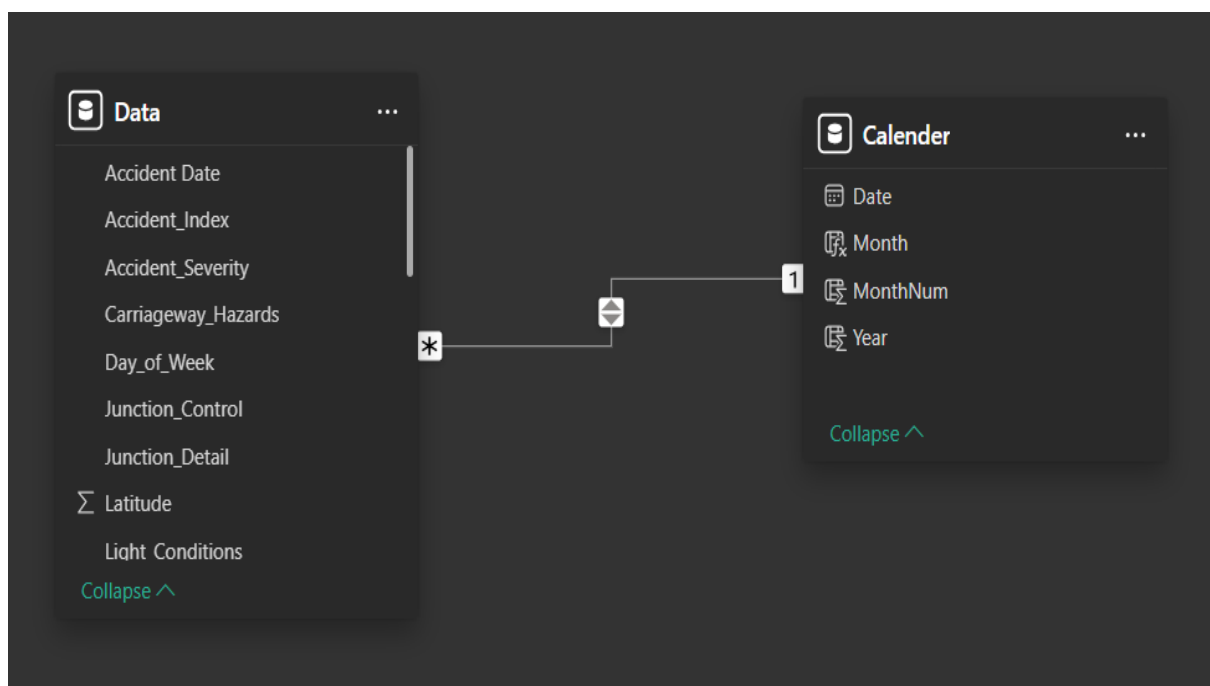


Figure 1: Data Modelling

- Imported dataset from the provided Google Drive link.
- Deleted 'Time' column which was irrelevant to the data.
- Corrected mislabelled data 'Fetal' to 'Fatal'.
- Cleaned missing or inconsistent values (especially for Carriageway Hazards).
- Converted date fields into proper date hierarchy (Year, Month, MonthNum, Day) into new table 'Calender'.

- Created DAX measures for custom KPIs and filtered statistics.
 - Accidents by Vehicle Type
 - Casualties
 - Total Accidents
 - Most Dangerous Weather Measure
 - High Speed Zone Accidents
 - Most Accident District name
 - Most Common Road Type for Accidents
 - Most Common Vehicle Type - Accidents
 - Urban Accident Percentage
 - Wet Road Causing Accidents

3 Main Dashboard – Road Accident Analysis



Figure 2: Main Dashboard - Road Accident Analysis (pg 1)

Purpose: Gives a holistic view of accident severity, distribution, and time-based trends.

Key Highlights:

- Total Accidents: 308K
- Casualties: 196K
- District with Most Accidents : Birmingham
- High Speed Zone Accidents (Speed greater than or equal to 60): 69K

Key Visuals:

- Monthly trend of accidents.
- Accident severity breakdown (Fatal, Serious, Slight).
- Accidents by day of week and urban/rural distribution.

- KPIs showing total accidents, casualties, and fatalities.
- Top Junction Controls Contributing To Accidents.
- Slicer for Area , Year and Weather Condition.
- Index Tab for Navigation Keys (Main Dashboard , Spatial and Environmental Analysis, Accident Trends by Road and Vehicle)
- Link for Database, Statistics and data about reported collisions and casualties on public roads in Great Britain.
- A button For Resetting all the Slicers.

4 Spatial and Environmental Analysis

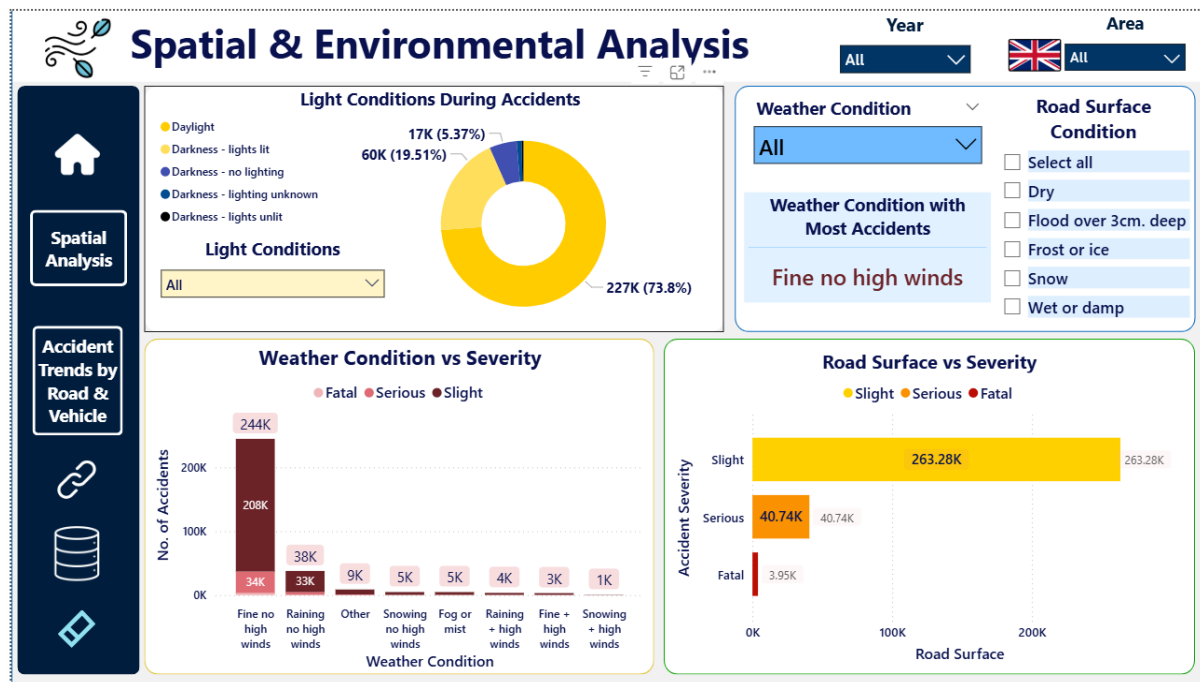


Figure 3: Spatial and Environmental Analysis (pg 2)

Purpose: This page explores how environmental and lighting conditions influence accidents.

Key Highlights:

- Most Common Weather Condition: Fine no high winds
- Light Condition with Most Accidents: Daylight
- Wet/Damp Road Accidents in Percentage: 26.56 percent of total

Key Visuals:

- Light Condition Donut Chart: Dominated by daylight accidents.
- Weather Condition vs Severity: Fine weather still shows highest accident volume.
- Road Surface vs Severity: Dry and wet roads linked to varying severities.
- Filters Used: Weather, Road Surface Condition

5 Accident Trends by Road and Vehicle

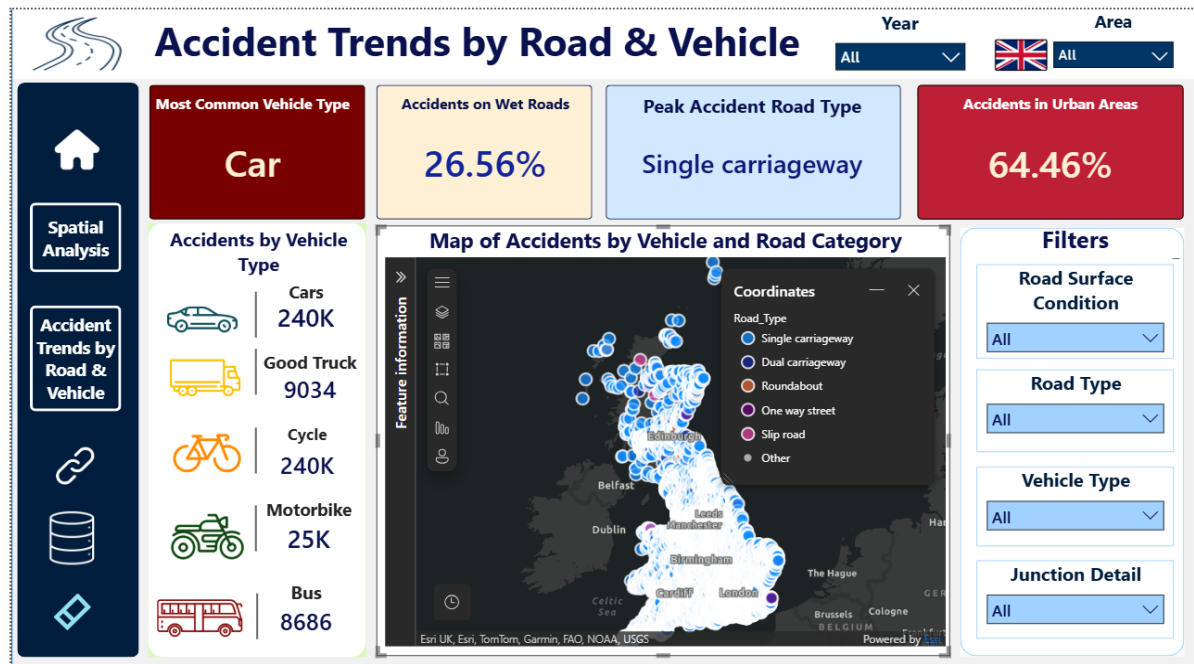


Figure 4: Accident Trends by Road and Vehicle (pg 3)

Purpose: Focused on analyzing road types and vehicle categories involved in accidents.

Key KPI Cards

- Most Common Vehicle in Accidents: Car
- Percentage Accidents on Wet Roads: 26.56 Percent
- Peak Accident Road Type: Single carriageway
- Urban Area Accident Share: 64.46

Visual Insights:

- Map Visualization: Interactive view of accident distribution across road categories.
- Vehicle Type Breakdown: Cars and cycles dominate the stats.
- Filters: Road Surface, Road Type, Vehicle Type, Junction Detail

6 Insights

- **Urban-Centric Interventions:** With over 64 percent of accidents occurring in urban areas, cities must be the primary focus for safety measures. This includes deploying speed cameras, redesigning junctions, and enhancing pedestrian crossings.
- **Infrastructure Improvements for High-Risk Roads:** Accidents are most frequent on single carriageways. Road widening, barrier installation, and adding service lanes may reduce collision risk on such roads.
- **Data-Driven Signage and Speed Control:** Areas with a speed limit of 60 or higher report more accidents. Enforcement of speed limits and predictive signage in high-speed corridors is critical.
- **Junction Design and Control:** A significant proportion of accidents occur at uncontrolled or poorly marked junctions. Investing in roundabouts, signals, and reflective signage can reduce incident frequency at these sites.
- **Vehicle-Type Targeted Campaigns:** Cars are the most involved vehicle category, but cycles and two-wheelers also show high vulnerability. Targeted road safety awareness programs, especially for urban commuters and delivery vehicles, are necessary.
- **Temporal Pattern Response:** Accident rates peak on Fridays and in the late months of the year. Temporary enforcement intensification (like checkpoints or patrols) during weekends and festive months may help mitigate surges.
- **Environmental Condition Sensitization:** Although most accidents occur in fine weather and daylight, accidents on wet roads and under fog/mist are often more severe. Real-time warnings and AI-based variable message signs can improve driver caution under adverse conditions.
- **Use of Smart Spatial Data for Urban Planning:** The dashboard includes spatial mapping of accident-prone districts. These hotspots can be flagged for future policy, signal timing optimization, and zoning decisions.

- **Public Transport Promotion:** Encouraging mass transit options may reduce the number of private vehicles, especially cars, which dominate accident statistics. Investment in BRTS, metro, or e-bike programs could have indirect safety benefits.
- **Holistic Monitoring with Power BI:** The dashboard model allows continuous and adaptive monitoring. It can be scaled for real-time data to support municipal dashboards and smart city control rooms for rapid response and forecasting.

7 Reference

- Road safety statistics
- Reported Road Casualties Great Britain
- Police.UK , Collisions and incidents